Alicia Garrido Peña, Ph.D.

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Personal details

Name	Alicia Garrido Peña
Current activity	Assistant Professor in Computer Engineering. Research member of Biological Neurocomputational Group (GNB)
Nationality	Spain
Affiliation	Departamento de Ingeniería Informática. Universidad Autónoma de Madrid.

Academic Background

2014 – 2018	**Computer Science Engineering** University of Granada.
2018 – 2019	**M.Sc. Computer Science Engineering** Autonomous University of Madrid (UAM).
	M.Sc. Research and Development in Information Technologies. UAM.
2020 – 2024	**PhD in Computer Engineering and Telecomunications.** ***Cum laude*****. International mention** UAM.

Internacionalization

September - December, 2021	**Research Stay**, University of Sussex, Brighton, United Kingdom. Professor: Thomas Nowotny.
December 1st-9th, 2025	**Teaching Stay**, Universidad Politécnica Salesiana, Quito y Cuenca, Ecuador.

Skills

Electrophysiology	Dynamic clamp, intracellular (sharp electrodes) and extracellular recordings. Expertise experience in *Lymnaea stagnalis* preparation and recording and *Carcinus maenas* STG recording.
Optical neurotechnology	Expertise in Infrared Laser stimulation.
Computational neuroscience	Conductance-based models, single cells and small circuits. High experience implementing and simulating.
Real-time and closed-loop	Design and performance of experiments with hard real-time combining living and simulated neurons.
Data analysis	High experience in data analysis with Python. Basic formation in fMRi with FSL
Coding	Python, C++, Java, Kotlin, SQL, XML/XSL, PHP, LaTeX
Languages	Strong Spanish (native) and English. Basic German.
Dissemination and Science Communication	Experience in oral presentations in conferences. High interest in science communication and dissemination.

Research Publications

Journal Publications

1. L. Ellison, G. Raiser, A. Garrido-Peña, G. Kemenes, and T. Nowotny, "SSSort 2.0: A semi-automated spike detection and sorting system for single sensillum recordings," *Journal of Neuroscience Methods*, p. 110 351, Dec. 2024, JRC Q2., ISSN: 0165-0270. DOI: 10.1016/j.jneumeth.2024.110351.

2. A. Garrido-Peña, P. Sanchez-Martin, M. Reyes-Sanchez, *et al.*, "Modulation of neuronal dynamics by sustained and activity-dependent continuous-wave near-infrared laser stimulation," *Neurophotonics*, vol. 11, no. 2, p. 024 308, May 2024, JCR Q1., ISSN: 2329-423X, 2329-4248. DOI: 10.1117/1.NPh.11.2.024308.

3. A. Garrido-Peña, I. Elices, and P. Varona, "Characterization of interval variability in the sequential activity of a central pattern generator model," *Neurocomputing*, vol. 461, pp. 667–678, Oct. 2021, JCR Q2., ISSN: 0925-2312. DOI: 10.1016/j.neucom.2020.08.093.

Conference Proceedings

1. A. Garrido-Peña, P. Sanchez-Martin, I. Elices, *et al.*, "Time precise closed-loop protocols for non-invasive infrared laser neural stimulation," in *CNS*, 2025.

2. A. Garrido-Peña, P. Sanchez-Martin, L. Ming, *et al.*, "Relationship between power, wavelength, delivery mode, and effect in infrared-laser neural stimulation.," in *SENC*, SENC meeting 2025, 2025.

3. A. Garrido-Peña, P. Sanchez-Martin, I. Elices, *et al.*, "Exploring the ability of biophysical models to reproduce the functional variability of neurons. 32nd Annual Computational Neuroscience Meeting: CNS*2023," en, in *Journal of Computational Neuroscience*, vol. 52, Oct. 2024, pp. 3–166. DOI: 10.1007/s10827-024-00871-5.

4. A. Garrido-Peña, P. Sanchez-Martin, M. Reyes-Sanchez, *et al.*, "Continuous-wave near-infrared laser stimulation non-invasively modulates neural dynamics in sustained and activity-dependent modalities," in *SFN*, SFN, Neuroscience 2024, 2024.

5. A. Garrido-Peña, P. Sanchez-Martin, M. Reyes-Sanchez, *et al.*, "Effective noninvasive neuronal waveform modulation with sustained and activity-dependent continuous-wave near-infrared laser stimulation," in *FENS*, Science Communications World Wide, 2024. DOI: 10.57736/6F07-365F.

6. A. Garrido-Peña, P. Sánchez-Martín, M. Reyes-Sanchez, *et al.*, "Activity-dependent infrared laser stimulation to assess its biophysical effects on single neurons. 31st Annual Computational Neuroscience Meeting: CNS*2022. F3. ORAL PRESENTATION.," in *Journal of Computational Neuroscience*, vol. 51, Jan. 2023, pp. 3–101. DOI: 10.1007/s10827-022-00841-9.

7. P. Sánchez-Martín, A. Garrido-Peña, B. Berbel, F. B. Rodriguez, R. Levi, and P. Varona, "Influence of electrical coupling in shaping time intervals and dynamical invariants of central pattern generator sequences. 31st Annual Computational Neuroscience Meeting: CNS*2022. P109.," in *Journal of Computational Neuroscience*, vol. 51, Jan. 2023, pp. 3–101. DOI: 10.1007/s10827-022-00841-9.

8. P. Soëtard, R. Amaducci, P. Sánchez-Martín, *et al.*, "Dynamical principles of functional neural sequences validated in hybrid robots built with living central pattern generators. 31st Annual Computational Neuroscience Meeting: CNS*2022. P110," in *Journal of Computational Neuroscience*, vol. 51, Jan. 2023, pp. 3–101. DOI: 10.1007/s10827-022-00841-9.

9. A. Garrido-Peña, P. Sanchez-Martin, R. Levi, J. Castilla, J. Tornero, and P. Varona, "Activity-dependent stimulation to assess the effect of infrared-laser stimulation in single neurons. Poster Presentation," in *FENS*, FENS forum 2022, Jun. 2022, S02–556.

10. R. Amaducci, I. Elices, M. Reyes-Sanchez, *et al.*, "Controlling robotic locomotion by a closed-loop interaction with living central pattern generators.," in *COSYNE*, Feb. 2021.

11 B. Berbel, A. Garrido-Peña, I. Elices, R. Latorre, and P. Varona, "Effect of Electrical Synapses in the Cycle-by-Cycle Period and Burst Duration of Central Pattern Generators," in *Advances in Computational Intelligence*, I. Rojas, G. Joya, and A. Català, Eds., ser. Lecture Notes in Computer Science, Springer International Publishing, 2021, pp. 81–92, ISBN: 978-3-030-85099-9. DOI: 10.1007/978-3-030-85099-9_7.

12 B. Berbel, A. Garrido-Peña, I. Elices, R. Latorre, and P. Varona, "Gap junctions shape the intervals that build robust sequences in a central pattern generator model. 30th Annual Computational Neuroscience Meeting: CNS*2021–Meeting Abstracts. P194.," in *Journal of Computational Neuroscience*, vol. 49, Dec. 2021, pp. 3–208. DOI: 10.1007/s10827-021-00801-9.

13 A. Garrido-Peña, I. Elices, R. Levi, F. B. Rodriguez, and P. Varona, "Universality of interval variability constraints in the sequential activity of motor circuits.," in *COSYNE*, Feb. 2021.

14 A. Garrido-Peña, R. Levi, J. Castilla, J. Tornero, and P. Varona, "Experimental and modeling study of near infrared-laser stimulation in single and electrically coupled neurons. Poster presentation.," in *SENC*, SENC meeting 2021, Nov. 2021.

15 A. Garrido-Peña, P. Sanchez-Martin, R. Levi, J. Castilla, J. Tornero, and P. Varona, "Effect of infrared laser stimulation in single neurons: Experimental and modeling study. 30th Annual Computational Neuroscience Meeting: CNS*2021–Meeting Abstracts. P193.," in *Journal of Computational Neuroscience*, vol. 49, Dec. 2021, pp. 3–208. DOI: 10.1007/s10827-021-00801-9.

16 P. Sanchez-Martin, I. Elices, A. Garrido-Peña, R. Levi, F. B. Rodriguez, and P. Varona, "Dynamic synchronization between electrically coupled cells of central pattern generators. 30th Annual Computational Neuroscience Meeting: CNS*2021–Meeting Abstracts. P195.," in *Journal of Computational Neuroscience*, vol. 49, Dec. 2021, pp. 3–208. DOI: 10.1007/s10827-021-00801-9.

17 R. Amaducci, I. Elices, M. Reyes-Sanchez, *et al.*, "Hybrid robot driven by a closed-loop interaction with a living central pattern generator with online feedback. 29th Annual Computational Neuroscience Meeting: CNS*2020. P207.," in *BMC Neuroscience*, vol. 21, Dec. 2020, p. 54. DOI: 10.1186/s12868-020-00593-1.

18 A. Garrido-Peña, I. Elices, R. Levi, F. B. Rodriguez, and P. Varona, "Experimental and computational characterization of interval variability in the sequential activity of the Lymnaea feeding CPG. 29th Annual Computational Neuroscience Meeting: CNS*2020. O11. ORAL PRESENTATION," in *BMC Neuroscience*, vol. 21, Dec. 2020, p. 54. DOI: 10.1186/s12868-020-00593-1.

Books and Chapters

1 A. Garrido-Peña, *TFM 2019/20. Máster en Ingeniería Informática*. TFM, Mar. 2022.

2 A. Garrido-Peña, *TFM 2019/20.Máster en Investigación e Innovación en Tecnologías de la Información y las Comunicaciones*. TFM, Mar. 2022.

Courses and activities

Neuroscience

February 2024 — **fMRi Postpfrocessing Course: Resting State & FSL**, Organized by the Center for Clinical Neuroscience. Hospital los Madroños. **Hours**: 12 hours.

October 2021 — **Hackathon Brain Code Games**, Organized by the Spanish Society of Neuroscience (SENC) and the Universidad Autónoma de Madrid (UAM). **Hours**: 60 hours.

Courses and activities (continued)

October 2020 — **First Neutouch Summer School: Touch for Prosthetics**, Online. Organizer: Neutouch.

April 2020 — **BCI & Neurotechnology Spring School 2020**, Online. Organizer: g.tec.

June 18, 2019 — **Biological Risk (4h)**, Madrid. Organizer: Prevention of Risks Service, Universidad Autónoma de Madrid (UAM).

July 4, 2019 — **Security on Chemicals Handling (4h)**, Madrid. Organizer: Prevention of Risks Service, Universidad Autónoma de Madrid (UAM).

Science dissemination and interdisciplinarity

March, 2024 and 2025 — **Organizationa and Conduction of course titled: "Free Software for Research Carreer"**, Universidad Autónoma de Madrid (UAM).

January 25, 2023 — **Conference "Open Science in the Spanish University and its Impact on Research Development and Evaluation"**, Interuniversity Institute INAECU.

May 2021 — **#HiloTesis Dissemination Contest**, Organized by CRUE.

October 2020 to February 2021 — **Interdisciplinary Views in Scientific Research (5 ECTS)**, Madrid. Organizer: EDUAM, Universidad Autónoma de Madrid (UAM).

February to June 2018 — **Scientific Entertainer Course at Science Museum (100h; 4 ECTS)**, Granada. Organizer: Parque de las Ciencias; University of Granada (UGR).

Women in Science and STEM

June 2024 — **AI Workshop for campus "I want to be an woman engineer"**, Universidad Autónoma de Madrid (UAM).

February 2024 — **Gender-Perspective Science Communication Course**, Science Culture Unit, Universidad Autónoma de Madrid (UAM).

June 2022 — **Participation in the Roundtable "Studying STEM"**, Project: "I Want to Be an Engineer - UAM", Organized by the School of Engineering. Universidad Autónoma de Madrid (UAM).

Teacher training courses

May 9–11, 2022 — **Towards an Inclusive University: Transformative Elements**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: — **Hours**: 6.

Courses and activities (continued)

February 15–17, 2023 — **How to Evaluate Learning Outcomes**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

April 20–27, 2023 — **Detection and Management of Psychological Issues Among Students**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

October 17–24, 2023 — **The Concept of Competence and Competency-based Training**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

February 19–20, 2024 — **Authority and Leadership Course**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

May–June 20–10, 2024 — **Design and Planning of the Teaching-Learning Process in the Context of a Course. Preparation of the Course Syllabus**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 2 **Hours**: 16.

October 18–25, 2024 — **Gamification and Game-based Learning: A Practical Workshop for Creating Educational Video Games**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

October 10–23, 2024 — **Flipping the Classroom: An Introduction to the Flipped Classroom**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

May 26–29, 2025 — **Evaluating Learning**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 12.

May–June 13–04, 2025 — **Creating a Dynamic Classroom**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 12.

October 10, 2025 — **Workshop "Teaching Blindly" (Fourth Edition)**, Faculty of Psychology, Universidad Autónoma de Madrid (UAM). **Event:** Accessible Psychology: "Without Groping in the Dark." **Hours**: 2.

October-November 24-7 — **Online Teaching and Intellectual Property: Practical Aspects for University Faculty**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 9.

November 4-11 — **Motivation and University Learning**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

Courses and activities (continued)

Other courses

February 2021 - June 2021 — **Organization and Participation in Research Group Seminars**, Organized by GNB.

June 2023 — **Participation in the Poster Exhibition at the Doctoral Week 2023**, Organized by the Graduate School. UAM.

September 2023 — **Attendance at the International Congress IBRO**, Organized by the International Brain Research Organization.

Academic Period 2017/2018 — **University of Granada Library: Formative Actions (3 ETCS)**, Granada. Organizer: University of Granada (UGR).

Grants and other merits

December 2025 — **Erasmus+ KA-171**, Grant for teaching mobility.

2025 — **Travel Award CNS*2025 conference**, Organization for Computational Neuroscience.

Travel Award SENC 2025 conference, Sociedad Española de Neurociencia.

June 2022 — **Thesis in 3 minutes Competition (honorable mention)**, Universidad Autónoma de Madrid, Alcalá de Henares, Politécnica de Madrid, Rey Juan Carlos, and Universidad Complutense.

2018-2019 — **Competitive grant for master studies in Universidad Autonoma de Madrid**, Master research program. Universidad Autónoma de Madrid.

2016-2017 — **Erasmus+**, University of Granada.

2016 — **Best Academic Certificate Award from E.T.S. de Ingenierías Informática y de Telecomunicación**, University of Granada.

Teaching experience

2020 to the present — Teaching experience in Computer Science Engineering, Telecommunication Engineering and Biomedical Engineering, and Master in Deep Learning for Audio and Video Processing. Courses related to Basic Programming, Operating Systems, Automata and Languages, Android programming, Deep Learning Basics.

Teaching experience (continued)

Courses Taught

Academic Year: 2019–2020

- **Course:** Programming II
  Degree Program: Bachelor's Degree in Computer Engineering
  Hours: 30h (1 lab section)
  Term: Second semester

- **Course:** Fundamentals of Computer Systems
  Degree Program: Bachelor's Degree in Telecommunications Technologies and Services Engineering
  Hours: 15h (1 lab section)
  Term: Second semester

Academic Years: 2020–2021, 2021–2022

- **Course:** Programming II
  Degree Program: Bachelor's Degree in Telecommunications Technologies and Services Engineering
  Hours: 60h (2 lab sections)
  Term: Second semester

Academic Years: 2022–2023, 2023–2024

- **Course:** Automata and Formal Languages
  Degree Program: Bachelor's Degree in Computer Engineering
  Hours: 60h (2 lab sections)
  Term: First semester

Academic Years: 2024–2025, 2025–2026

- **Course:** Operating Systems
  Degree Program: Bachelor's Degree in Computer Engineering
  Hours: 60h (2 lab sections)
  Term: Second semester

Academic Year: 2024–2025

- **Course:** Project Administration and Management
  Degree Program: Bachelor's Degree in Biomedical Engineering
  Hours: 30h (1 lecture section)
  Term: Second semester

Academic Years: 2024–2025, 2025–2026

- **Course:** Mobile Application Development
  Degree Program: Bachelor's Degree in Computer Engineering
  Hours: 60h (project-based course)
  Term: Second semester

- **Course:** Deep Learning Fundamentals and Basic Tools
  Degree Program: Master's Degree in Deep Learning for Audio and Video Processing
  Hours: 12h
  Term: First semester

Teaching experience (continued)

Short Courses and Workshops

March 4 and 11, 2024

Course: Free Software for Research Careers
Autonomous University of Madrid (UAM)

June 2024

Course: AI Workshop for the "I Want to Be an Engineer" Program
Autonomous University of Madrid (UAM)